H2 Geography Paper 2 Suggested Answers

Section A
Cluster 4 Fieldwork

1 A group of 12 students decided to investigate the flood risk in Bangkok, Thailand. More specifically, they were keen to understand the threat experienced by people in different flood risk zones. They visited fifteen households in each flood risk zone over three days—Monday, Wednesday, and Friday—and interviewed one member from each household using a questionnaire. The results of this questionnaire were then compared across the three flood risk zones.

Resource 1 shows the flood risk zones in Bangkok, Thailand. Resource 2 shows the questionnaire used in the investigation. Resource 3 shows the results of Question 6c in the questionnaire compared between the three flood risk zones.

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(a) Suggest a suitable hypothesis for the fieldwork investigation and state two reasons why it is at a suitable scale. [3]

Award 1 mark for suitable hypothesis for the fieldwork investigation
Award 1 mark for each reason why the investigation is at a suitable scale, to maximum of 2 marks.

- Hypothesis: *"Residents living in higher flood risk zones in Bangkok are more likely to experience severe flood-related threats."* [1 mark]
- The study is conducted at a local scale within Bangkok, allowing manageable comparison of three distinct zones by a group of 12 students [1 mark].
- The scope focuses on perception and impact at household level, which is appropriate for student-led data collection using questionnaires [1 mark].

(b) With reference to Resource 1, explain the benefits of collecting data at the three sites shown. [4]

Award 1 mark for each explanation of the data collecting benefits, to maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

- The three sites represent a gradient of flood risk—high, moderate, and low—which allows for a comparative analysis of spatial variation in flood threats [1 mark].
- Coverage of different land uses (e.g., areas near levees, industrial plants, and airport zones) offers varied contexts for interpreting threat perception [1 mark].
- Including the Chao Phraya River floodplain and levee-protected canal areas enables students to assess how proximity to engineered defences influences perceptions [1 mark].
- Data collection across these zones allows the identification of patterns and anomalies, enhancing the analytical depth of the study [1 mark].

(c) Suggest two ethical issues students should consider when conducting this investigation. [5]

Award 1 mark for each suggestion of ethical issues when conducting the investigation, to a maximum of 5 marks.
Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

- Informed consent: Students must explain the purpose of the study and obtain verbal or written agreement from participants before starting the questionnaire to ensure participants willingly contribute data [1 mark].
- Confidentiality and anonymity: Personal information such as names and addresses should not be recorded or disclosed. Responses should be kept anonymous to protect the privacy and dignity of the individuals surveyed [1 mark].

- Right to withdraw: Participants should be informed that they can stop the interview at any point or refuse to answer specific questions without any consequences or pressure [1 mark].
- Avoidance of psychological distress: Questions about flood damage to homes or belongings (e.g., Question 6c in Resource 2) may trigger traumatic memories. Students must be sensitive in their tone, phrasing, and response to emotional reactions [1 mark].

(d) Suggest how students can improve their sampling for the investigation. [6]

Award 1 mark for each suggestion on how to improve the sampling for the investigation, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

- Adopt stratified sampling within each zone: Divide each flood risk zone into smaller sub-areas (e.g., near levees vs further inland) and ensure proportional representation from each, improving coverage and reducing location bias [1 mark].
- Increase sample size beyond 15 households per zone: A larger sample size (e.g., 25–30 per zone) enhances statistical reliability and reduces the influence of outliers or unrepresentative responses [1 mark].
- Implement systematic random sampling: Rather than choosing households by convenience, students could survey every nth house (e.g., every 5th house along a street), reducing bias in selection and increasing objectivity [1 mark].
- Survey at different times of the day and week: Including weekends and evenings allows students to reach working individuals who may not be available during weekday hours, improving respondent diversity [1 mark].

(e) Explain the strengths and limitations of the questionnaire shown in Resource 2. [6]

Award 1 mark for each explanation of a strength or a limitation.
Award a maximum of 1 additional mark for further development of each explanation.
Award a maximum of 2 marks for explanation of either strength or weakness only.

Strengths:

- Questions such as 6c use a Likert scale, allowing quantifiable data for analysis and comparison across zones [1 mark].
- The inclusion of open-ended Q5 provides qualitative insights into residents' comparative perceptions beyond their own zone [1 mark].
- The structure includes both objective and subjective elements, covering past experiences and future expectations [1 mark].

Limitations:

- Some questions (e.g., Q2) rely on respondent's personal understanding of flood risk, which may be inaccurate or uninformed [1 mark].
- The questionnaire lacks socio-demographic data (e.g., age, income, household size), limiting the ability to identify vulnerable groups [1 mark].
- Questions on impacts (e.g., 6a–6h) may produce overlapping or ambiguous interpretations without clear definitions of terms (e.g., "services") [1 mark].

(f) Using Resource 3, explain what the results suggest about the threat experience of the people living in each of the three zones. [6]

Award 1 mark for each explanation of the threat experience, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

- All 5 respondents from the high-risk zone rated flood threat to internal house/property as "1 – Very unlikely", which contradicts the official classification of high flood risk and suggests a very low perceived threat despite the high hazard zoning [1 mark].
- This perception may be influenced by visible or effective mitigation infrastructure, such as the levee near Khlong Prapa or engineered defences in Ayutthaya, which instills confidence in flood protection [1 mark].
- All respondents in the moderate-risk zone gave no ratings at either extreme ("5 – Very likely" or "1 – Very unlikely"), which may indicate uncertainty or low awareness of flood threats in their area, reflecting a lack of direct experience or exposure to flood events [1 mark].
- In the low-risk zone, one respondent rated the likelihood of internal flooding as "5 – Very likely", while two others rated it as "1 – Very unlikely", suggesting a wide variation in threat perception among residents despite being in the same risk zone [1 mark].

(g) Evaluate the effectiveness of this investigation in understanding the threat experienced by people in different flood risk zones. [10]

Possible Approaches:

Candidates could refer to Resource 3 to assess how well the questionnaire results reflect actual flood threat. Although the Likert-scale responses allow for quantifiable comparison, findings such as all high-risk zone respondents selecting "very unlikely" and one low-risk respondent choosing "very likely" suggest a disconnect between perceived and official flood risk (Resource 1). This limits the investigation's ability to accurately capture flood threat experiences.

Candidates may also consider limitations in sampling and data design. The small sample size (15 per zone), absence of demographic data, and weekday-only collection reduce representativeness and reliability. The questionnaire (Resource 2) captures perception but lacks triangulation with physical data such as past flood events, elevation, or drainage conditions.

Despite these weaknesses, candidates could acknowledge strengths such as the clear spatial comparison across zones and use of structured perception indicators. However, to improve effectiveness, more comprehensive data and triangulation would be needed.

Levels marked using Generic Level Descriptors for H2 fieldwork evaluative questions.

Levels	Marks	Generic Level Descriptors for H2 Fieldwork Evaluative Question
3	9–10	Evaluation is analytical and coherent. Response addresses the question and demonstrates good knowledge and understanding of fieldwork methods relevant to the given context.
2	5–8	Evaluation is mostly analytical and coherent. Weaker responses in this level will have evaluation that is broadly analytical and generally coherent. Response generally addresses the question and demonstrates adequate knowledge and understanding of fieldwork methods relevant to the given context.
1	1–4	Response is descriptive with limited or no analysis and evaluation. Response is fragmented and lacks clarity. Response lacks focus on the question and may be largely irrelevant to it. Response shows basic knowledge and understanding of fieldwork methods. Response has some, though limited, relevance to the given context.
0	0	No creditworthy response

Section B

2 Resource 4 shows the number of slum population in Mumbai district, India in 2020. Resource 5 shows access to basic amenities in urban and rural India in 2024. Resource 6 shows a slum housing site in Mumbai, India.

(a) Suggest two relevant indicators to monitor sustainable urban development. [4]

Award 1 mark for each suggestion of relevant indicators to monitor SUD.
Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

1. Proportion of urban households with access to improved sanitation facilities (e.g., flush toilets or improved latrines) [1 mark]:
   This indicator reflects the social well-being dimension of SUD as it signals progress in public health, hygiene standards, and quality of life in urban areas [1 mark].
2. Percentage of urban population using clean cooking fuel (e.g., LPG, electricity, biogas) [1 mark]:
   This measures environmental integrity, as clean fuels reduce indoor air pollution and reliance on unsustainable biomass, improving both environmental quality and health outcomes [1 mark].
3. Unemployment rate in urban areas [1 mark]:
   This indicates the level of economic vitality, as high employment levels are crucial for sustaining livelihoods, generating income, and supporting inclusive urban economic growth [1 mark].
4. Percentage of urban households with access to piped water within dwelling [1 mark]:
   This is an indicator of social well-being and environmental integrity, showing the availability and accessibility of basic services that improve health, convenience, and reduce the burden on marginalised groups, especially women [1 mark].

(b) Describe the distribution of slum population in Mumbai District as shown in Resource 4. [4]

Award 1 mark for each description of distribution of slum population, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each description, where applicable.

- The slum population in Mumbai District in 2020 is spatially uneven, with a higher concentration in central and eastern districts and lower concentrations in peripheral areas [1 mark].
- Red-shaded districts (326,236–658,972), such as Ghatkopar, Kurla, and Andheri East, show the highest slum populations, reflecting areas with dense informal settlements and proximity to key transport and industrial hubs [1 mark]. Orange-shaded districts (201,592–326,235), such as Chembur and Dadar, also experience high slum populations, often adjacent to the highest-density zones [1 mark].
- Light blue districts (0–60,893), such as Colaba to Metro and Byculla, located towards the southern peninsula, show lower slum populations [1 mark].
- Some light green (60,894–151,506) and yellow (151,507–201,591) districts such as Goregaon and Mulund [1 mark] are interspersed between high- and low-density zones [1 mark].

(c) Compare the access to basic amenities in urban India and rural India, as shown in Resource 5. [5]

Award 1 mark for each comparison of access to basic amenities in urban and rural India, to a maximum of 5 marks.

Award a maximum of 1 additional mark for further development of each comparison, where applicable. Maximum 3 marks for either similarities or differences.

Similarities:

- High access to mobile phones in both areas:
  Mobile usage is widespread, with 92% in rural areas and 95% in urban areas, indicating strong telecommunication infrastructure nationwide [1 mark].
- High reported access to improved latrines:
  Both urban (98%) and rural (95%) populations report high access, suggesting success in sanitation outreach across regions [1 mark].
- Moderate access to primary drinking water within dwellings:
  While there is a gap, both rural (58%) and urban (82%) households have access to drinking water within their dwelling, reflecting progress in water infrastructure development [1 mark].

Differences:

- Access to piped water into dwellings:
  Urban households have significantly higher access (61%) compared to rural areas (24%), indicating better infrastructure provision in cities [1 mark].
- Internet usage gap:
  Internet access is much higher in urban areas (64%) than rural areas (40%), highlighting a digital divide that limits rural access to digital services and opportunities [1 mark].
- Access to clean cooking fuel:92% of urban households use clean cooking fuel versus only 50% in rural areas, showing greater environmental health risks and energy poverty in rural settings [1 mark].

(d) Citing data from Resource 5, explain why rural populations continue to move into urban areas in countries at low level of development. [5]

Award 1 mark for each explanation using data from Resource 5 on why rural populations continue to move into urban areas in less developed countries.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

- Better access to piped water in urban areas:
  In urban areas, 61% of households have access to piped water into dwellings compared to only 24% in rural areas [1 mark]. This access to basic services motivates rural households to migrate for improved living standards and convenience [1 mark].
- Improved sanitation infrastructure in urban settings:
  98% of urban residents report access to improved latrines, whereas rural areas have slightly lower access at 95% [1 mark]. While the gap may appear small, the quality and reliability of urban sanitation services tend to be higher, offering greater hygiene and dignity [1 mark].
- Greater access to clean cooking fuel in urban areas:
  92% of urban households use clean cooking fuel, in contrast to just 50% in rural households [1 mark]. Rural populations are exposed to higher health risks due to indoor air pollution from traditional fuels, encouraging migration to cities with cleaner energy sources [1 mark].
- Urban digital connectivity provides more opportunities:
  Urban areas have 64% internet penetration compared to 40% in rural areas [1 mark]. This digital divide limits access to information, education, and economic opportunities in rural areas, pushing younger populations toward urban centres [1 mark].

- Urban living perceived as offering better prospects:
  As seen in Resource 5, the combination of better infrastructure, healthcare access, job opportunities, and connectivity in cities reinforces the perception that urban areas provide better life chances, contributing to continued rural-urban migration [1 mark].

(e) With reference to Resource 6, explain how the multiple deprivations experienced by slum dwellers in Mumbai can impact sustainable urban development. [6]

Award 1 mark for each explanation on how the multiple deprivations experienced by slum dwellers can impact SUD, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

- Inadequate sanitation and waste management affects environmental integrity [1 mark]:
  Resource 6 shows widespread litter and stagnant water in front of slum dwellings. This reflects poor waste disposal and drainage systems, which can cause waterborne diseases and environmental pollution, undermining environmental goals of SUD [1 mark].
- Overcrowded and poorly built housing compromises social well-being [1 mark]:
  The tightly packed, makeshift structures seen in the image indicate overcrowding and insecure housing. This leads to health and safety risks, weakens community resilience, and violates the principle of inclusive and safe urban living [1 mark].
- Lack of access to clean water limits health and hygiene [1 mark]:
  Visible water containers and buckets in Resource 6 suggest unreliable or absent piped water access. This limits the ability of slum residents to maintain hygiene, contributing to disease outbreaks that strain urban healthcare systems [1 mark].

(f) Explain why slums may also develop in urban areas of countries at higher level of development. [6]

Award 1 mark for each explanation on why slums may also develop in urban areas of developed countries, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

- Housing affordability crisis due to high urban land and rental prices [1 mark]:
  In cities like San Francisco (USA) and London (UK), skyrocketing real estate prices and rent increases have made housing unaffordable for many low- to middle-income residents. As a result, individuals are pushed into informal housing such as overcrowded shared flats or makeshift shelters [1 mark].
- Migrant and undocumented populations may be excluded from formal housing systems [1 mark]:
  In Rome, Italy, many undocumented migrants live in abandoned buildings or informal camps due to lack of access to formal rental markets. Similarly, Los Angeles (USA) has seen the rise of informal settlements by migrant workers in marginalised urban spaces [1 mark].
- Urban poverty and income inequality persist even in wealthy nations [1 mark]:
  In New York City (USA), more than 60,000 people are homeless on any given night. Many others live in slum-like conditions in public housing that is severely under-maintained, indicating that economic inequality leads to informal or inadequate housing even in high-income settings [1 mark].

Section C

Cluster 3: Sustainable Future and Climate Change

3 ‘Environmental factors pose the greatest challenge to liveability in urban areas.’

Evaluate this statement. [20]

Possible Approaches:

Candidates could assess whether environmental factors — such as air and noise pollution, poor waste management, and lack of green spaces — are the greatest challenge to liveability, due to their direct impact on health, well-being, and quality of life. They may use examples like Delhi’s poor air quality or Zurich’s environmental success to support their argument. Alternatively, candidates may argue that socio-economic challenges (e.g. poverty, inadequate housing, poor access to services) or social inequalities (e.g. gender-based safety concerns, ageing populations) pose more persistent and fundamental threats to liveability.

Candidates could also take a contextual approach, showing that what challenges liveability most depends on the place, time, and social group being assessed. For instance, basic needs may take precedence in low-income cities, while inclusiveness and psychological well-being may matter more in high-income contexts. Stronger responses will evaluate the relative importance of different factors, using case studies and acknowledging the subjective, dynamic, and multidimensional nature of liveability.

Levels marked using Generic Level Descriptors for 20m H2 essays

4 Evaluate the view that contemporary climate change is attributed more to natural factors than human activities. [20]

Possible Approaches:

Candidates could evaluate the extent to which natural processes or human activities account for contemporary climate change. They may consider natural factors such as solar variability, volcanic eruptions, and ocean circulation, which have historically influenced climate but operate over longer timescales or cause short-term fluctuations. In contrast, recent rapid warming aligns more closely with anthropogenic activities—especially fossil fuel use and deforestation—which have increased greenhouse gas concentrations.

Candidates could reference scientific consensus, particularly the IPCC, and highlight how climate models only match observed trends when human-induced emissions are included. They may also apply geographical concepts like time (recent vs. historical change), scale (regional vs. global), and system (feedback loops intensified by human actions).

A well-argued response should weigh both perspectives but is likely to conclude that human activities play a greater role in driving contemporary climate change.

Levels marked using Generic Level Descriptors for 20m H2 essays

Levels	Marks	Generic Level Descriptors for H2 Essays
5	18–20	Evaluation is consistently analytical and coherent. Response is well-supported by relevant material, including the effective use of examples. Response features accurate geographical knowledge and reflects good understanding of the subject content relevant to the question.
4	14–17	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	10–13	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	6–9	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–5	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response

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